DNS & DHCP Server Setup: Project Documentation

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Tool: Cisco Packet Tracer,VS code,Github

1. Introduction

The company operates a single local area network (LAN). Until now, every new employee computer needed a manually assigned static IP address, which is slow and error-prone. Employees also had to remember the exact IP address of the internal web server just to open the intranet.

This project solves both problems by deploying two network services:

- **DHCP (Dynamic Host Configuration Protocol)** to hand out IP settings to computers automatically.
- **DNS (Domain Name System)** so the intranet can be opened with a friendly name, `intranet.company.local`, instead of an IP address.

2. Objectives

- Remove the need to configure IP addresses manually on employee PCs.
- Allow users to reach the intranet by name instead of by IP address.
- Keep servers and the gateway on fixed addresses that never change.
- Document the design, configuration and testing so it can be repeated.

3. Scope and Requirements

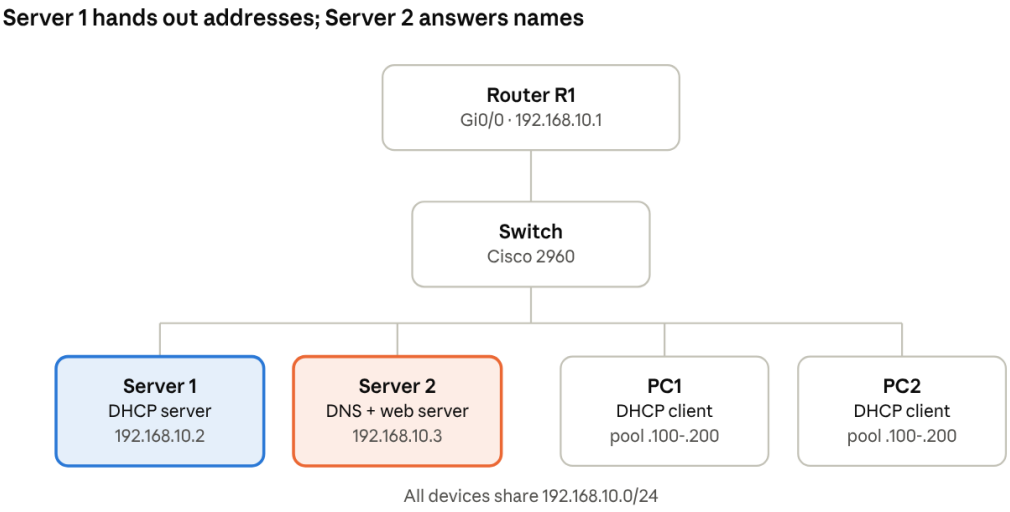
Devices

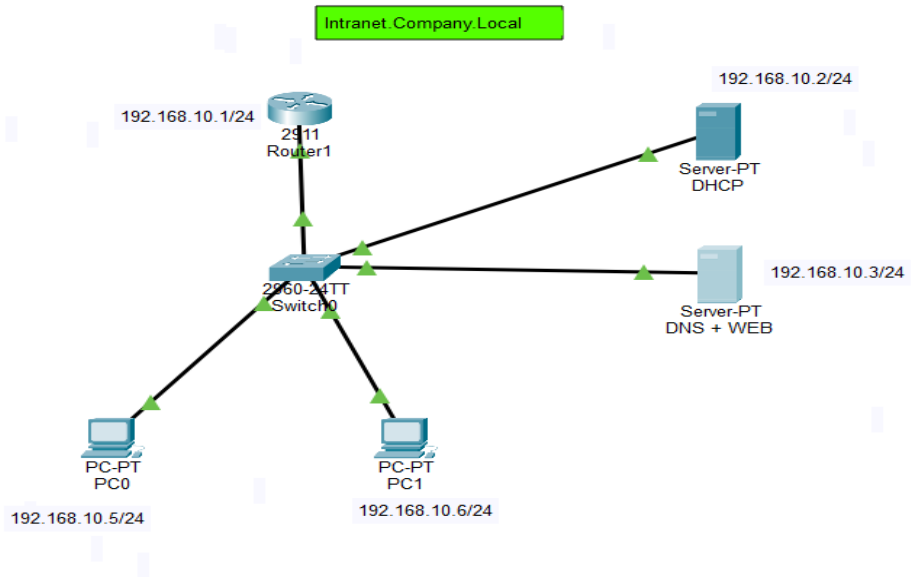
Quantity	Device	Model (Packet Tracer)	Purpose
1	Router	Cisco 2911	Default gateway for the LAN
1	Switch	Cisco 2960	Connects all devices
1	Server 1	Server-PT	DHCP server
1	Server 2	Server-PT	DNS server and intranet web server
2	PCs	PC-PT	Employee computers (DHCP clients)

4. Network Design

4.1 Topology

The router connects to the switch through its GigabitEthernet0/0 interface. Server 1, Server 2, PC1 and PC2 each connect to the switch with copper straight-through cables. All devices share one subnet.





Network topology · 1 router, 1 switch, 2 servers, 2 PCs

Server 1 and Server 2 keep fixed addresses; both PCs take theirs from the DHCP pool.

4.2 Cabling

From	To	Cable
Router Gi0/0	Switch Gi0/1	Copper straight-through
Switch Fa0/1	Server 1	Copper straight-through
Switch Fa0/2	Server 2	Copper straight-through
Switch Fa0/3	PC1	Copper straight-through
Switch Fa0/4	PC2	Copper straight-through

4.3 IP addressing plan

Device	Interface	IP address	Subnet mask	Assignment
Router R1	Gi0/0	192.168.10.1	255.255.255.0	Static
Server 1 (DHCP)	NIC	192.168.10.2	255.255.255.0	Static
Server 2 (DNS/Web)	NIC	192.168.10.3	255.255.255.0	Static
PC1	NIC	192.168.10.100 to 200	255.255.255.0	DHCP
PC2	NIC	192.168.10.100 to 200	255.255.255.0	DHCP

Addresses 192.168.10.1 to 192.168.10.10 are reserved for infrastructure devices, so the DHCP pool never hands them out.

4.4 DHCP pool settings

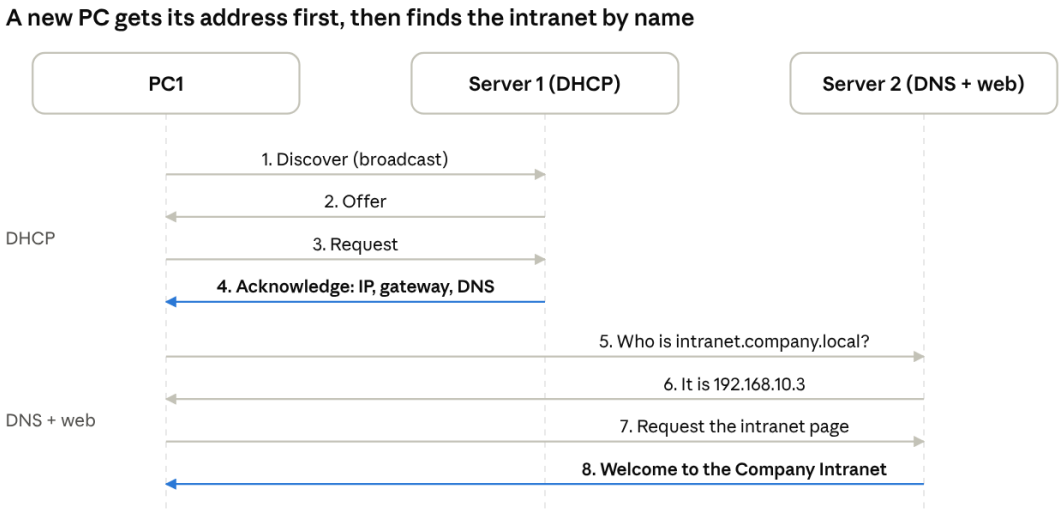
Setting	Value
Pool name	LAN_POOL
Network	192.168.10.0/24
Start address	192.168.10.100
Maximum users	101 (192.168.10.100 to 192.168.10.200)
Default gateway	192.168.10.1
DNS server	192.168.10.3

4.5 DNS records

Name	Type	Address
intranet.company.local	A	192.168.10.3

5. How the Solution Works

1. A new PC is connected and set to obtain an address automatically.
2. The PC broadcasts a DHCP request. Server 1 answers with an IP address, subnet mask, default gateway and DNS server (the DHCP four-step process: Discover, Offer, Request, Acknowledge).
3. The user opens a browser and types `http://intranet.company.local`.
4. The PC asks the DNS server (192.168.10.3) to resolve the name. Server 2 replies with 192.168.10.3.
5. The browser connects to the HTTP service on Server 2 and the intranet page loads.



working of DHCP, DNS and HTTP · 8 messages

Messages 1 to 4 are the DHCP lease; messages 5 to 8 are the DNS lookup and the web page.

6. Implementation

Step 1. Place the devices

Open Packet Tracer and add one 2911 router, one 2960 switch, two Server-PT devices and two PC-PT devices to the canvas.

Step 2. Connect the cables

Use copper straight-through cables as listed in section 4.2. Wait until all link lights are green. The router link turns green after the interface is enabled in Step 3.

Step 3. Configure the router

Open the router, select the CLI tab and enter

```
enable
configure terminal
hostname R1
```

```
interface gigabitEthernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
```

Step 4. Configure Server 1 as the DHCP server

1. Open Server 1, then Desktop, then IP Configuration. Choose Static and enter IP 192.168.10.2, mask 255.255.255.0, gateway 192.168.10.1, DNS 192.168.10.3.
2. Open the Services tab, select DHCP and turn the service On.
3. Enter the pool details from section 4.4 (name LAN_POOL, gateway 192.168.10.1, DNS 192.168.10.3, start IP 192.168.10.100, mask 255.255.255.0, maximum users 101).
4. Click Save.

Step 5. Configure Server 2 as the DNS and web server

1. Open Server 2, then Desktop, then IP Configuration. Choose Static and enter IP 192.168.10.3, mask 255.255.255.0, gateway 192.168.10.1, DNS 192.168.10.3.
2. Open Services, then DNS, and turn the service On. Add the name `intranet.company.local`, type A Record, address 192.168.10.3, and click Add.
3. Open Services, then HTTP, and turn the service On. Edit `index.html` so it displays "Welcome to the Company Intranet", and save.

Step 6. Configure the client PCs

On PC1 and PC2, open Desktop, then IP Configuration, and select DHCP. After a few seconds each PC should show an address in the 192.168.10.100 range.

7. Testing and Verification

No.	Test	How to run it	Expected result	Result
1	DHCP lease on PC1	IP Configuration set to DHCP	Address 192.168.10.100 or higher, gateway .1, DNS .3	Pass
2	DHCP lease on PC2	IP Configuration set to DHCP	A different address from PC1 in the same range	Pass
3	DHCP server shown	`ipconfig /all` on PC1	DHCP server listed as 192.168.10.2	Pass
4	Gateway reachable	`ping 192.168.10.1`	Replies received	Pass
5	DNS resolution	`nslookup intranet.company.local`	Returns 192.168.10.3	Pass
6	Ping by name	`ping intranet.company.local`	Replies from 192.168.10.3	Pass
7	Intranet	Web Browser,	Intranet welcome page loads	Pass

No.	Test	How to run it	Expected result	Result
	access	`http://intranet.company.local`		
8	Add a new PC	Connect a third PC set to DHCP	Receives the next free address automatically	Pass

8. Results

After configuration, new computers receive their IP address, subnet mask, gateway and DNS server automatically with no manual input. The intranet is reachable by the name `intranet.company.local`, and employees no longer need to know the web server's IP address. The infrastructure devices keep fixed addresses, so the network stays predictable.

9. Conclusion

This project shows how DHCP and DNS remove two everyday administrative burdens on a small LAN. DHCP on Server 1 assigns network settings automatically, and DNS on Server 2 lets users open the intranet by name. The result is a simpler, more reliable network that is easy to expand as the company grows.